AI/ML Internship: Week 4 Practical Assignment

1. Core Objective

Welcome to the final week of your internship! In Week 3, you optimized, tuned, and validated your machine learning model. However, a model sitting inside a Python notebook cannot serve real-world business users.

Your goal for Week 4 is **Deployment & Production Readiness**. You will extract your saved model from your notebook, wrap it into an interactive Web UI or REST API endpoint (using frameworks like Streamlit, Flask, or FastAPI), host it, write production documentation, and present your work in the Final Internship Showcase.

2. Project Tracks & Deployment Guidelines

You will deploy the optimized model you refined during Week 3 using one of the following deployment pathways:

Track Category	Week 3 Optimized Component	Week 4 Final Deployment Tasks
Track A: Tabular Prediction Framework (Regression)	Tuned Regularized Regression Model (Ridge/Lasso).	**Serialization:** Export model artifact (`.pkl` or `.joblib`).**Deployment:** Build a Streamlit or Flask UI allowing users to input numerical attributes via forms/sliders to receive real-time value predictions.
Track B: Textual Classification Pipeline (NLP)	Optimized Text Classifier with TF-IDF Vectorizer.	**Serialization:** Export both the TF-IDF Vectorizer and model object.**Deployment:** Create a web interface with a text area where users can type custom sentences/reviews and view instant classification tags.
Track C: Automated Visual Recognition (Computer Vision)	Optimized Image Classification Model.	**Serialization:** Save pre-processing pipeline and trained neural/KNN weights.**Deployment:** Build a drag-and-drop web portal (Streamlit/Gradio) that accepts user

		image uploads and displays predicted visual labels.

3. Technology Stack & Required Tools

- **Model Serialization:** `joblib` or `pickle` (To package and export trained models out of notebooks).
- **Web Framework Options:**
    - **Option 1 (Recommended for Data Science UI):** Streamlit or Gradio (100% Python-based interactive frontend).
    - **Option 2 (Recommended for Backend Engineering):** FastAPI or Flask (For REST API endpoints).
- **Free Hosting Platforms:** Streamlit Community Cloud, Hugging Face Spaces, or Render.
- **Version Control:** GitHub Repository (Mandatory to house code, model files, and `requirements.txt`).

4. Step-by-Step Deployment Workflow

1. **Model Exporting:** Save your best model and transformers from Week 3 as binary files (e.g., `model.pkl`).
2. **App Script Creation:** Write a clean Python file (`app.py` or `main.py`) that loads the binary model, handles input validation, passes inputs through pre-processing, and generates predictions.
3. **Environment Packaging:** Create a `requirements.txt` file listing all necessary Python packages and version numbers.
4. **GitHub Repository Setup:** Push your app code, saved model, and dependency list to a public GitHub repository.
5. **Cloud Deployment:** Link your GitHub repo to Streamlit Community Cloud, Hugging Face Spaces, or Render to generate a publicly accessible live web app link.

5. Final Review & Viva Deliverables

To successfully graduate from the internship, you must submit two final items prior to your Final Viva Showcase:

Deliverable 1: Production Documentation (`README.md`)

A complete `README.md` file located inside your GitHub repo containing:

- **Project Overview:** Title, summary, and business problem solved.
- **Live App Link:** A functional, clickable URL to your deployed web application/API.
- **System Architecture:** Description of how data flows from user input $\rightarrow$ app interface $\rightarrow$ pre-processing pipeline $\rightarrow$ saved model $\rightarrow$ output.
- **Installation & Setup Instructions:** Step-by-step terminal commands (`git clone`, `pip install`, `streamlit run`) to run your app locally.

Deliverable 2: Final Showcase Presentation Blueprint (6-Slide Structure)

A final presentation deck (Max 5-minute showcase + 2-minute live Q&A/Viva) structured as follows:

- **Slide 1: Title & Executive Summary:** Project name, track, intern details, and final live link.

- **Slide 2: Problem Statement & Dataset:** The core challenge solved and overview of the dataset used.
- **Slide 3: Technical Evolution (Week 1 to Week 3):** Brief summary of model selection, feature engineering, tuning, and accuracy gains achieved over time.
- **Slide 4: Deployment Architecture:** Flowchart or diagram showing how your app loads `.pkl` files and serves user queries.
- **Slide 5: Live App Demonstration:** Live walk-through or GIF showing the app accepting inputs and displaying predictions.
- **Slide 6: Lessons Learned & Future Improvements:** Real-world challenges faced during deployment and prospective next steps.